BIO401_BIOSTATISTICS

Past Mid Terms

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Subjective Questions

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1. **Types of Sampling**

   Sampling types include **Simple Random Sampling** (each member has an equal chance), **Stratified Sampling** (population divided into subgroups or strata with proportional samples), **Systematic Sampling** (samples taken at regular intervals, such as every nth individual), and **Cluster Sampling** (population divided into clusters, and entire clusters are randomly chosen). Each method improves data representativeness depending on the population structure.

2. **Systematic and Cluster Probability**

   Systematic Sampling involves selecting every nth element, which is efficient but may introduce bias if there is a periodic pattern. **Cluster Sampling** involves dividing the population into clusters and randomly selecting some clusters, studying all elements within selected clusters. This method is efficient for geographically spread populations.

3. **Comparison between Population and Sampling**

   A **population** includes all individuals or items of interest, while a **sample** is a representative subset. Sampling is essential when studying an entire population is impractical. Populations yield parameters (e.g., population mean), while samples provide statistics (e.g., sample mean) that approximate the parameters with a margin of error.

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4. **Branches of Statistics**

Statistics branches into **Descriptive** and **Inferential Statistics**. **Descriptive Statistics** summarize data (e.g., mean, median), while **Inferential Statistics** use sample data to draw conclusions about a population (e.g., hypothesis testing). Descriptive statistics offer insight into data patterns, while inferential statistics extend insights to broader populations.

5. **Quantitative and Qualitative Variables**

Variables can be **Quantitative** (numerical, measuring quantities, e.g., height) or **Qualitative** (categorical, describing attributes, e.g., color). Quantitative variables may be **discrete** (countable, e.g., number of students) or **continuous** (within a range, e.g., weight). Variable types determine the choice of statistical tests.

6. **Example of Nominal Scale**

Nominal scales classify data without a quantitative relationship. For example, nationality (e.g., American, Canadian) has categories with no inherent order. Nominal scales are used in qualitative data analysis to categorize and count observations.

7. **How We Select a Sample in Cluster Sampling**

In **Cluster Sampling**, a population is divided into clusters (e.g., by region), then a few clusters are randomly selected. All members within selected clusters are sampled. This method reduces cost for geographically dispersed populations but may introduce cluster bias if clusters differ significantly.

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8. **Formula of Variance**

Variance measures data spread by averaging squared deviations from the mean. The formula is:

Variance (σ^2) = $\Sigma(x - \bar{x})^2 / N$ where $\Sigma(x - \bar{x})^2$ represents the sum of squared deviations from the mean $\bar{x}$, and N is the total number of observations. Variance is fundamental for calculating standard deviation.

9. **Random and Biased Error**

Errors in data are either **Random** (unpredictable fluctuations) or **Biased (Systematic)**, where results skew in one direction due to methodological flaws. Controlling for bias is crucial for accurate and reliable results.

10. **Sampling Types and Why We Need Samples**

    Different sampling types (random, stratified, systematic, cluster) help create representative samples, which are essential when studying every population member is impractical. Samples allow researchers to infer conclusions about the population costeffectively.

11. **Types of Clusters**

    Clusters represent natural groupings within a population, such as regions or schools. In **Cluster Sampling**, entire clusters are randomly selected, and all individuals within chosen clusters are studied. Proper cluster selection helps ensure data representativeness.

12. **What is Data and Source of Data?**

    Data is information collected for analysis. **Primary data** is gathered firsthand (e.g., surveys), while **Secondary data** is from existing sources (e.g., publications). Primary data offers tailored insights, while secondary data is often more accessible.

13. **Types of Distribution**

    Common distributions include **Normal Distribution** (bell-shaped, symmetric), **Binomial Distribution** (two outcomes with fixed probabilities), **Poisson Distribution** (rare events over time intervals), and **Uniform Distribution** (equal probability for all outcomes).

14. **Box and Whisker Plot**

    A **box and whisker plot** displays data distribution, showing the median, quartiles, and outliers. The box spans Q1 to Q3, with whiskers extending 1.5 times the interquartile range (IQR), while outliers are plotted individually.

15. **Pie Chart**

    A **pie chart** is a circular chart divided into slices, with each slice representing a data proportion. It's useful for showing parts of a whole but is limited for detailed comparisons, best for categorical data with a few categories.

16. **Chebyshev's Inequality**

    Chebyshev's Inequality states that for any dataset, at least $(1 - 1/k^2)$ of data lies within k standard deviations of the mean, where $k > 1$. It's valuable for datasets that don't follow normal distribution, establishing bounds on data spread.

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17. **Definitions of Types of Sampling**

Sampling types include **Random Sampling** (equal selection chance), **Systematic Sampling** (fixed intervals), **Stratified Sampling** (population divided into subgroups with samples from each), and **Cluster Sampling** (whole groups sampled).

18. **Primary Data and Secondary Data**

Primary data is gathered directly by researchers, while **Secondary data** comes from pre-existing sources. Primary data provides specific insights but is costly, whereas secondary data saves resources but may lack specificity.

19. **Frequency Distribution and Frequency Histogram**

A **frequency distribution** is a table that shows data values and their frequencies, while a **frequency histogram** is a bar graph of this table, with intervals on the x-axis and frequencies on the y-axis.

20. **Steps of Whisker Plot**

To create a **box and whisker plot**, calculate Q1, Q3, and IQR. Draw a box from Q1 to Q3, plot the median inside, and extend whiskers 1.5 times the IQR. Points beyond whiskers are outliers.

21. **Outlier**

An outlier is a data point that significantly deviates from others. It lies beyond 1.5 times the IQR or outside a certain number of standard deviations from the mean. Outliers can affect calculations of mean and standard deviation.

22. **Biased or Standard Error Definition**

Biased error skews data consistently due to design flaws, while **Standard Error (SE)** measures the precision of a sample mean. $SE = s / \sqrt{n}$, where s is sample standard deviation and n is sample size. Lower SE indicates greater precision.

23. **Draw Histogram**

A histogram is a bar graph that represents the frequency distribution of a dataset. Data is divided into bins (intervals), and the height of each bar shows the frequency in each bin, useful for visualizing data spread and central tendency.

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24. **Assumption of Measurement Scale**

Measurement scales include **Nominal** (categories), **Ordinal** (ordered categories), **Interval** (ordered with equal intervals), and **Ratio** (ordered with true zero). Proper scale use ensures valid analysis, as each scale supports different statistics.

25. **Continuous and Discrete Variable**

Continuous variables can take any value within a range, like weight. **Discrete variables** represent distinct values, like number of students. Recognizing the type helps in selecting statistical techniques; for example, mean is suited for continuous data.

26. **Excel Sheet**

Excel sheets organize data into rows and columns, with statistical functions like AVERAGE(), STDEV(), and COUNTIF() for calculations. It's a versatile tool for data entry, analysis, and visualization through charts and formulas.

27. **Is It Easy to Derive a Sample from Homogeneous and Finite Population Than from Heterogeneous and Infinite Population?**

Sampling is easier from a **homogeneous, finite population** as it is consistent and limited, reducing complexity. In contrast, **heterogeneous, infinite populations** require complex techniques like stratified sampling for representativeness.

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28. **Probability**

Probability measures event likelihood, calculated as P(A) = Favorable outcomes / Total outcomes. Probability ranges from 0 (impossible) to 1 (certain) and is foundational in assessing risk and uncertainty in statistical models.

29. **Mean**

Mean (average) is the sum of all values divided by the number of values: Mean ($\bar{x}$) = $\Sigma x / n$ where Σx is the sum of all values, and n is the number of values. The mean represents central tendency but is sensitive to outliers.

30. **Median**

Median is the middle value in ordered data. For odd n, it's the middle value; for even n, it's the average of two central values. Median is robust to outliers, making it suitable for skewed data.

31. **Mode**

Mode is the most frequently occurring value in a dataset. There may be one mode (unimodal), multiple modes (multimodal), or none if all values are unique. Mode is useful for identifying common categories in nominal data

Objective Questions (Derived from Quiz & Mids)

1. **Statistics** is the art and science of drawing inferences/conclusions about a population of interest.
2. **Probability** for a continuous random variable to assume any particular value is zero.
3. We need sampling for **estimation** purposes.
4. Example of simple random sampling is: **lottery method**.
5. A **box plot** is a visual representation of the five-number summary of the distribution of a numerical variable.
6. If the random variable X can have only countable values x1, x2, ..., we say it has a **discrete distribution**.
7. One of the earliest known demographic/Bio-statistics studies was conducted by **John Graunt**.
8. **Mean** is affected by extreme values.
9. Probability sampling is divided into **four** types.
10. Which of the following is NOT a Characteristic of a good sample?

    Nonrepresentativeness.
11. If the p-value is 0.029 using alpha = 0.05 as the level of significance criterion, **reject the null hypothesis**.
12. **Microsoft Excel** is a computer program used to create electronic spreadsheets.
13. **Alternative Hypothesis** is the hypothesis believed to be true by the researcher.

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14. **Convenience sampling** is commonly known as unsystematic, accidental, or opportunistic sampling.
15. The **interquartile range (IQR)** is the difference between the upper quartile and the lower quartile.
16. Which one is not a type of Probability Sampling design? **Judgment sampling**.
17. **Individual frequency distribution** is used for datasets with few different values. Each value is in its own class.
18. **Relative frequency histogram** has the same shape and the same horizontal scale as the corresponding frequency histogram.
19. The drawing of inferences about a body of data when only part of the data is observed is called **statistical inference**.
20. **Mean** of values 115, 110, 119, 117, 121, and 126 is: **118**.
21. **Median** for the given values 23, 28, 28, 31, 32, 34, 37, 42, 50, 61 will be: **32**.
22. **Convenience sampling** is known as unsystematic, accidental, or opportunistic sampling.
23. A **box plot** is a visual representation of the five-number summary of a numerical variable.
24. **Class boundaries** are the smallest numbers that can actually belong to different classes.
25. If the random variable X can have only countable values x1, x2, ..., we say it has a **discrete distribution**.
26. An assumption about the population parameter is called a **statistical hypothesis**.
27. The average value of a data set is called the **mean**.
28. Probability sampling is divided into **four** types.
29. **Ronald Fisher** is known as the "Father of Modern Statistical Sciences."
30. “An aggregate of entities for which we have an interest at a particular time” is called a **population**.
31. **Snowball sampling** refers to identifying someone who meets the criteria for inclusion in the study.

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32. **Individual type of frequency distribution** is used for data sets with few different values.
33. A **sample** is the subset of a collection of all possible entities described by a parameter.
34. A **discrete variable** is a variable that can only take specific values.
35. The purpose of **stratified random sampling** is to ensure representation of all subgroups.
36. A sequence of Bernoulli trials forms a **Bernoulli process**.
37. Each Bernoulli trial results in one of the **two** possible outcomes.
38. The names of the two possible outcomes are **Failure and Success**.
39. The probability of success is denoted by **P**.
40. The probability of success remains **constant** from trial to trial.
41. The trials are **independent**.
42. A **binomial probability distribution** results from a Bernoulli process.
43. A binomial probability distribution has a **fixed** number of trials.
44. **P and q** will denote the probabilities of success (S) and failure (F), respectively.
45. "P" is the **probability of success**.
46. "q" is the **probability of failure**.
47. "n" represents a fixed number of **trials**.
48. "x" represents a **specific number of successes in "n" trials**.
49. P(x) is equal to the **probability of getting exactly "x" successes among the "n" trials**.
50. "x = " represents the **probability of "x" successes among "n"**.
51. There are **three** methods for finding probability.
52. In a **binomial distribution**, we find the probabilities corresponding to the random variable "x".
53. Methods of finding probability include **using the binomial probability formula, using the table of probabilities, and using technology**.
54. Suppose we have n = 5 and p = 0.60. The probability that binomial trials yield four successes is given by: **0.05184**.

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55. One has to be certain that both "x" and "p" refer to the same category being called a **success**. 56. When sampling is conducted without replacement, events are considered **independent** if $n < 0.05$.
57. The **range rule of thumb** suggests that values are unusual if they are outside of these limits.
58. $\mu + 2\sigma$ represents the **maximum value**.
59. $\mu - 2\sigma$ represents the **minimum value**.
60. For $p = 0.5$ and a large or small "n," the binomial distribution will be **symmetric**.
61. For a large "p" and small "n," the binomial distribution will be **left-skewed**.
62. For a small "p" and small "n," the binomial distribution will be **right-skewed**.
63. For a small "p" but large "n," the binomial distribution approaches **symmetry**.
64. In MS Excel, we use the function **BINOMDIST**.
65. The number of events occurring in a given unit of time is represented as **2**.
66. Commonly used measure of central tendency is **mean**.
67. Qualitative variables include **blood group**.
68. "An aggregate of entities for which we have an interest at a particular time" is called **population**.
69. Sampling problems may differ in each strata. This refers to **stratified random sampling**.
70. The source of data can be **all of the above**.
71. Procedure for Systematic Random Sampling includes the following except **randomly select a number from 0 to k**.
72. One of the earliest known demographic/Bio-statistics studies was conducted by **John Graunt**.
73. Population is the collection of all possible entities, which is described by **parameters**.
74. In NCn, N represents **the number of cases in the sampling frame**.
75. Probability sampling is divided into **6** types.

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76. Example of simple random sampling is **get list of clients over the past year**.
77. Ordinal data scale is used for measurement by **categories**.
78. Which of the following is NOT a characteristic of a good sample? **Informative with maximum use of resources**.
79. A descriptive value for a population is called **parameter**.
80. The average value of data is called **mean**.
81. Sometimes **stratified sampling** is also called "proportional" or "quota" random sampling.
82. If a clinic administrator wishes to obtain information regarding the mode of transportation used by patients, the study should be **survey**.
83. The type of sampling that involves identifying someone who meets the criteria for inclusion in the study is **snowball sampling**.
84. Median for the values 23, 28, 28, 31, 32, 34, 37, 42, 50, 61 is **31**.
85. The hypothesis that is believed to be true by the researcher is the **alternative hypothesis**.
86. Which one is not a type of Probability Sampling design? **Non-systematic sampling**.
87. In sampling design, "n" stands for **the number of cases in the sample**.
88. The hypothesis that is rejected by the researcher is **null hypothesis**.
89. We need sampling for the purpose of **all of the above**.
90. The mean of values 115, 110, 119, 117, 121, and 126 is **118**.

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